

Homework 4
Principles of Artificial Intelligence
7th June, 2023

Question 1.

Jerry is a gambler. He went on a trip to Las Vegas and decided to make a fortune by leveraging his deep knowledge he learned from textbooks. In the casino, there are three slot machines, and there are two possible outcomes for each pull: you either win (get 20 dollars) or lose (get 0 dollars). Considering the budget, Jerry can only play 1000 rounds. Please help Jerry with this.

Part 1: Maximizing the Reward (70pt)

- Set the probability of winning for the three slot machines to be 0.3, 0.7, and 0.8, respectively.
- Implement the following three algorithms: (i) ϵ -greedy; (ii) upper confident bound with $C = 0.5$; and (iii) Thompson sampling algorithm.
- For each algorithm, report the following three results: (i) the final accumulated reward; (ii) the final accumulated regret; and (iii) how many times each arm has been pulled.
- Plot the distribution after 1000 rounds. Note that for simplicity, you may use beta distribution in all three cases. An example is shown in Fig. 1.
- Discuss your observation on the differences among the three algorithms.

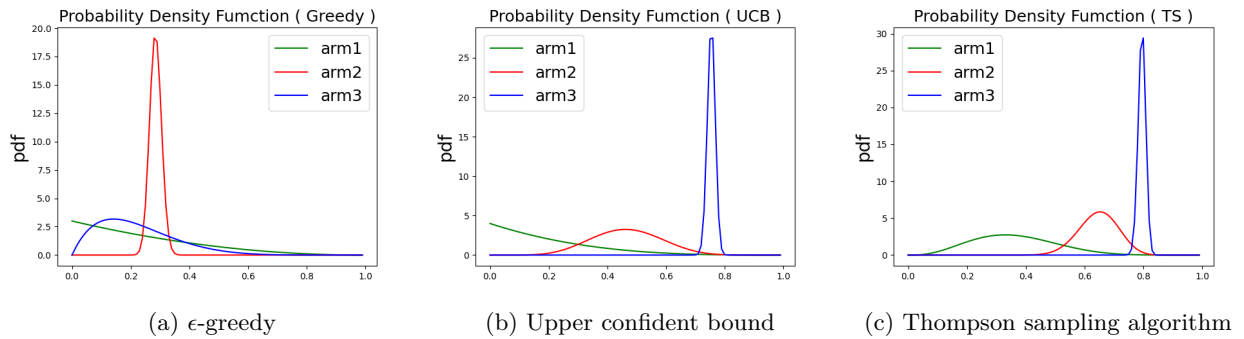


Figure 1: An example of the probability density functions for the three algorithms.

Part 2: Hyperparameter Selection (30pt)

For the previous part, play around with various settings and discuss your results.

- Upper confident bound:
 - You may change the parameter of C , the winning probability for each bandit, the round you can play, etc. You may also change the amount of arms.
 - Plot the confidence bound at round 100, 400, 700, 1000 and observe how it changes.
- Thompson sampling algorithm:
 - You may change the parameter of the winning probability for each bandit, the round you can play, etc. You may also change the amount of arms.
 - Plot the distribution in rounds 100, 400, 700, 1000 and observe how it changes. Note that, for simplicity, you may use the beta distribution in all three cases.

Hints:

- TAs are your friends. Please **DO NOT** hesitate to contact the TAs if you have questions.

Instruction: Please submit **three files** for this assignment:

1. Program File

Submit your source code file.

- File name format: HW4_XXXXXXXXXXXXXXXXXX.py or HW4_XXXXXXXXXXXXXXXXXX.zip
Replace XXXXXXXXXXXX with your student ID(s).
- Your program should be executable and reproducible.

2. Short Video (3 minutes)

Submit a short video demonstrating your implementation.

- File name format: HW4_XXXXXXXXXXXXXXXXXX.mp4
Replace XXXXXXXXXXXX with your student ID(s).
- Video length: **3 minutes**
- The video should include:
 - **First 1 minute:** Explain your code and demonstrate that the program runs correctly.
 - **Next 2 minutes:** Present and explain your experimental results.

3. Simulation Results (PDF)

Submit a PDF document containing your result figures.

- File name format: HW4_XXXXXXXXXXXXXXXXXX.pdf
Replace XXXXXXXXXXXX with your student ID(s).
- The PDF should contain organized figures of your simulation results.